1 The Limit of the Sequence of Terms of a Convergent Series

Theorem 1.1. *If the series* $\sum_{i=1}^{\infty} a_i$ *is convergent, then* $\lim_{i\to\infty} a_i = 0$.

1. Prove Theorem 1.1. (*Hint*: $a_i = s_i - s_{i-1}$.)

Remark 1.2. With any series $\sum a_i$ we associate two sequences: the sequence $\{s_n\}$ of its partial sums and the sequence $\{a_i\}$ of its terms. If $\sum a_i$ is convergent, then the limit of the sequence $\{s_n\}$ is s (the sum of the series) and, by Theorem 1.1, the limit of the sequence $\{a_i\}$ is 0.

2 The Harmonic Series

Is the converse to Theorem 1.1 true? That is, if $\{a_i\}$ is a sequence which converges to 0, does the series $\sum_{i=1}^{\infty} a_i$ converge? To investigate this question, consider the *harmonic series*

$$\sum_{i=1}^{\infty} \frac{1}{i} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots.$$

2. Calculate $\lim_{i\to\infty} a_i$ for the harmonic series.

To study the convergence of the harmonic series, it is useful to consider the partial sums $s_2, s_4, s_8, \ldots, s_{2^n}, \ldots$.

Proposition 2.1. *For the harmonic series,* $s_{2^n} > 1 + \frac{n}{2}$ *for* $n \geq 2$.

Proof. (By induction.)

3. To establish the base case, show that $s_4 > 1 + \frac{2}{2}$.

4. Write the inductive hypothesis.

5. Prove the inductive step to complete the proof.

□

6. Calculate $\lim_{n\to\infty} s_n$ for the harmonic series.

7. Is the converse to Theorem 1.1 true?

3 The Divergence Test

The contrapositive of Theorem 1.1 provides a test for divergence.

Theorem 3.1 (Divergence Test)**.**

(a) If $\lim_{n\to\infty} a_n \neq 0$*, then the series* $\sum_{n=1}^{\infty} a_n$ *diverges.*

(b) If $\lim_{n\to\infty} a_n = 0$*, then the test is inconclusive.*

Remark 3.2. The Divergence Test is also frequently called the *nth-term test*.

8. What does the Divergence Test say about each of the following series?

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{5n^2+4}$

(b) $\sum_{n=1}^{\infty} \arctan(n)$?

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{e^n}{n^2}$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{1+\left(\frac{2}{3}\right)^n}$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$